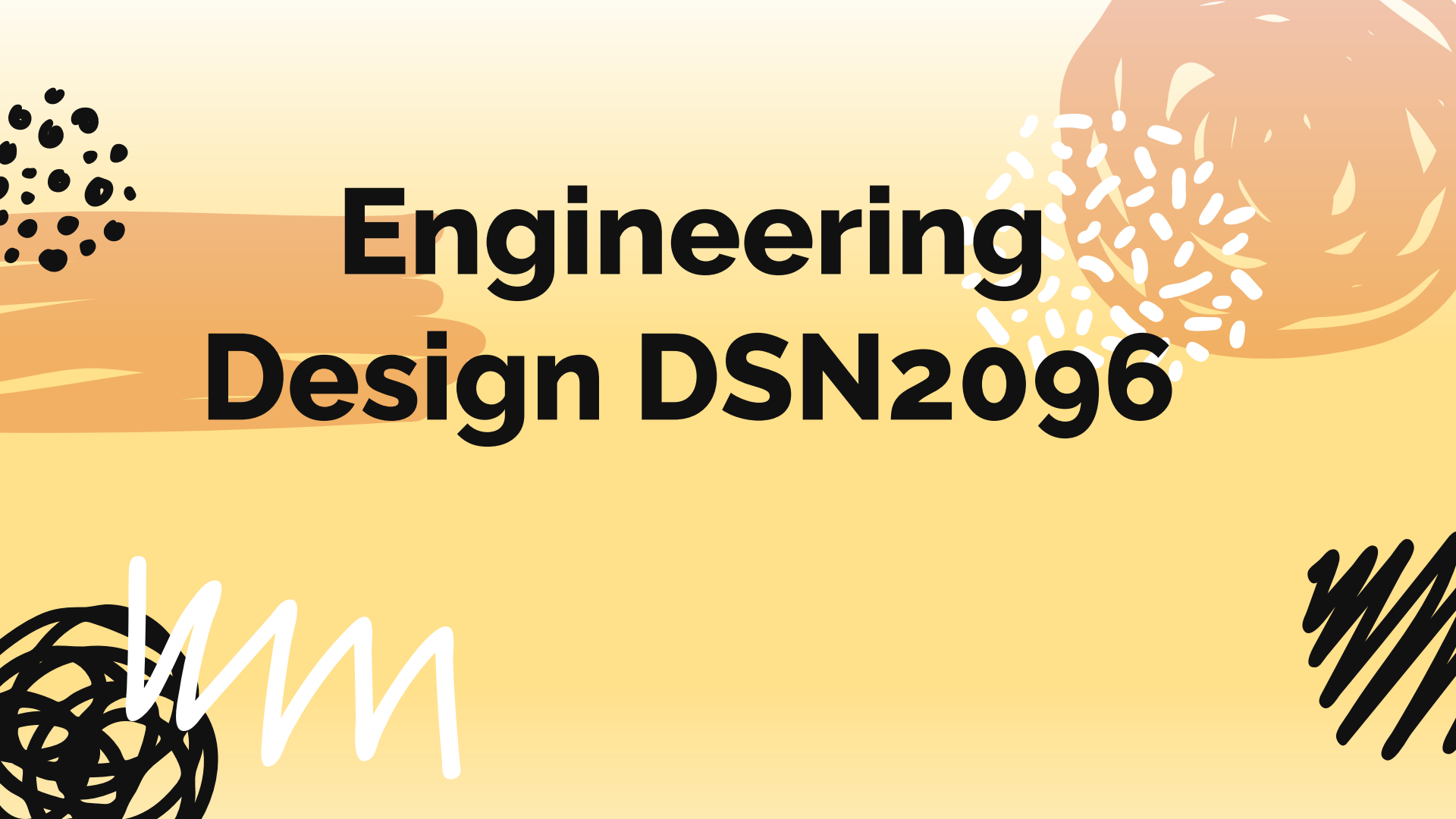




Engineering Design DSN2096

Syllabus



- **Introduction to Engineering design process:** Design History; Dieter Rams Principles of Good Design; Overview of Engineering Design Process: Problem Formulation, Concept generation, Project Planning and Design Making; Human Centred Design (HCD);
 - **Design Thinking for Innovation:** Design Thinking as Mindset, Process and Toolbox., Enhancing Design Thinking Through, Empathy, Interviewing, Questioning & Brainstorming, Tools for Design Thinking: Mind Mapping, Innovation Flowchart – Question ladder – SCAMPER (for products) Journey Mapping, Task analysis grid (for services)
 - **Engineering Design Approaches:** Professional and societal Context of Design; Different types of design – Conceptual, Embodiment designs and Detailed designs – Identification and Specifications, Standards and codes, Design Features – Design for Aesthetics, Production, Standards, Minimum risk, Ease of maintenance, Quality, Minimum cost and Optimum Design, Service Design - People - Asserts - policies - culture. Iterative process - Exploration - Creation - Reflection - Implementation.
- 

Syllabus



- **Usability & Reliability:** Usability – User requirement; User experience; Usability testing; Customer Co-creation; **Reliability & Safety:** Human and equipment, safety, Risk analysis, and security, System reliability; **Use of e-engineering:** Modelling, Simulation and Verification.
 - **Prototyping and Visualization:** Design Cycle Model, Metaphor method: Theory and methodology of concept generation, Blend method & Thematic Method. Conceptual Design & Design capability and sociality. Prototyping & Visualization Design Tools – E-tools
 - **Sustainable Design:** Concepts of sustainable development, Sustainable design principles - Design for Environment; Life Cycle Assessment; Models of sustainable design- Biomimicry, Eco Design, Recycling; Social Innovation
 - **Communication:** Articulating design ideas: Storytelling; Sketching & Dynamic Diagrams; K Scripts
- 



Books and resources

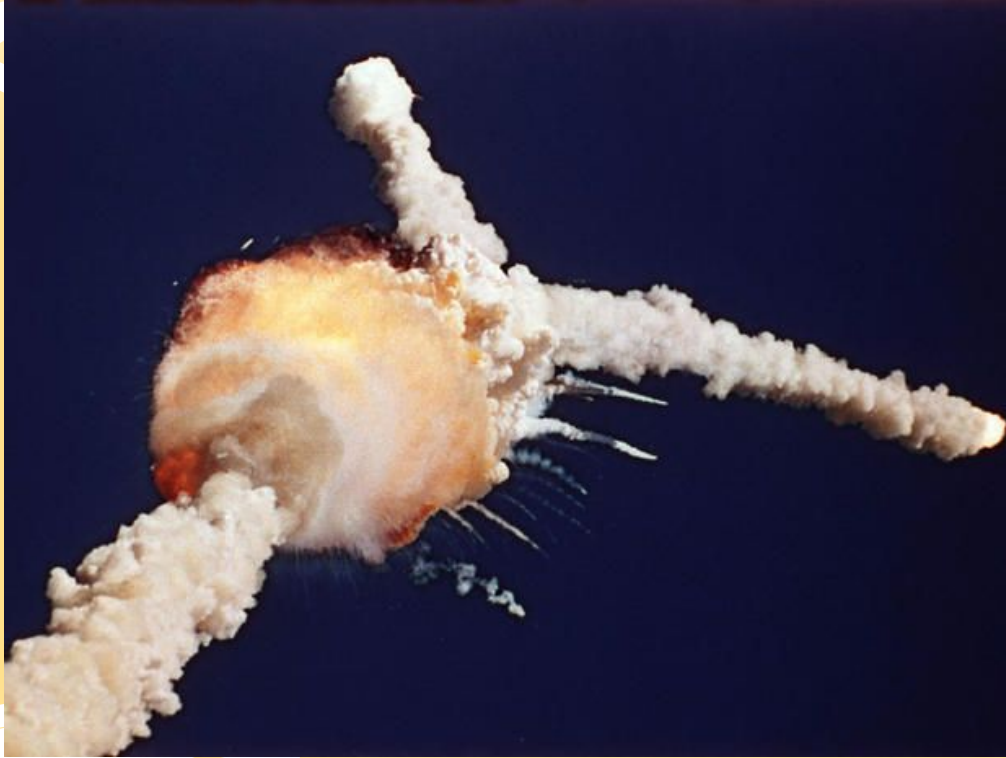
TEXT BOOKS:

1. Huge Jack, “Engineering Design, Planning, and Management” Academic Press, 2013.
2. Gerhard Pahl, Wolfgang Beitz “Engineering Design: A Systematic Approach” 2014.

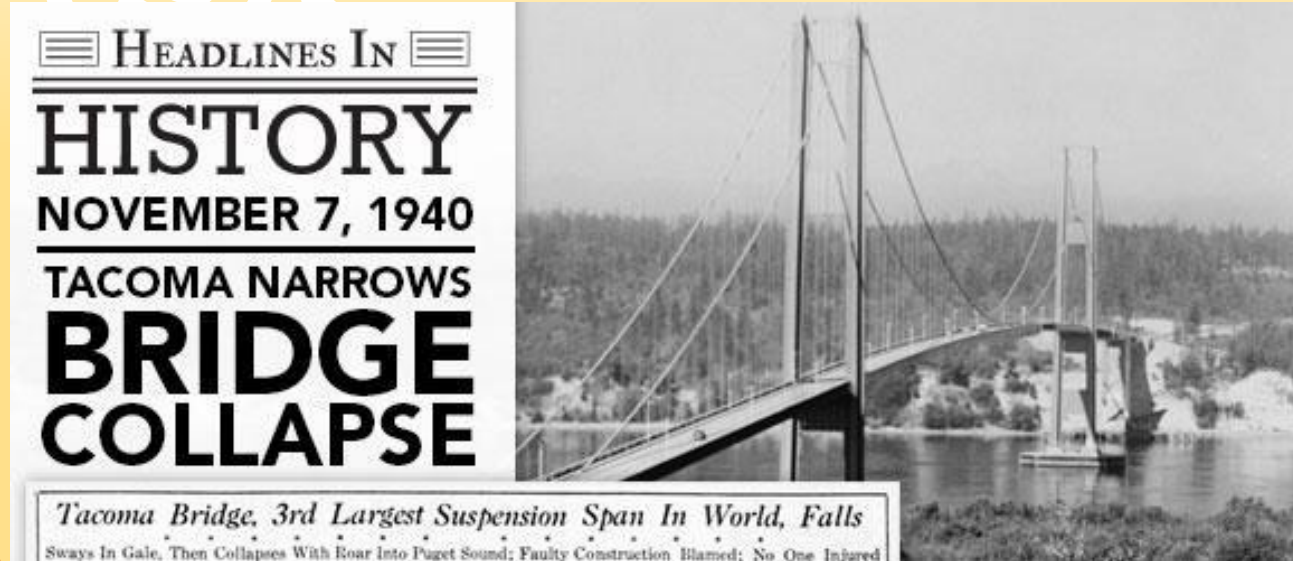


**Let's start with some
problems**

Challenger Disaster



Tacoma Bridge Collapse



Collapse of The Quebec Bridge



Bhopal Gas Tragedy





**Why do we need
good design?**

Design by Nature



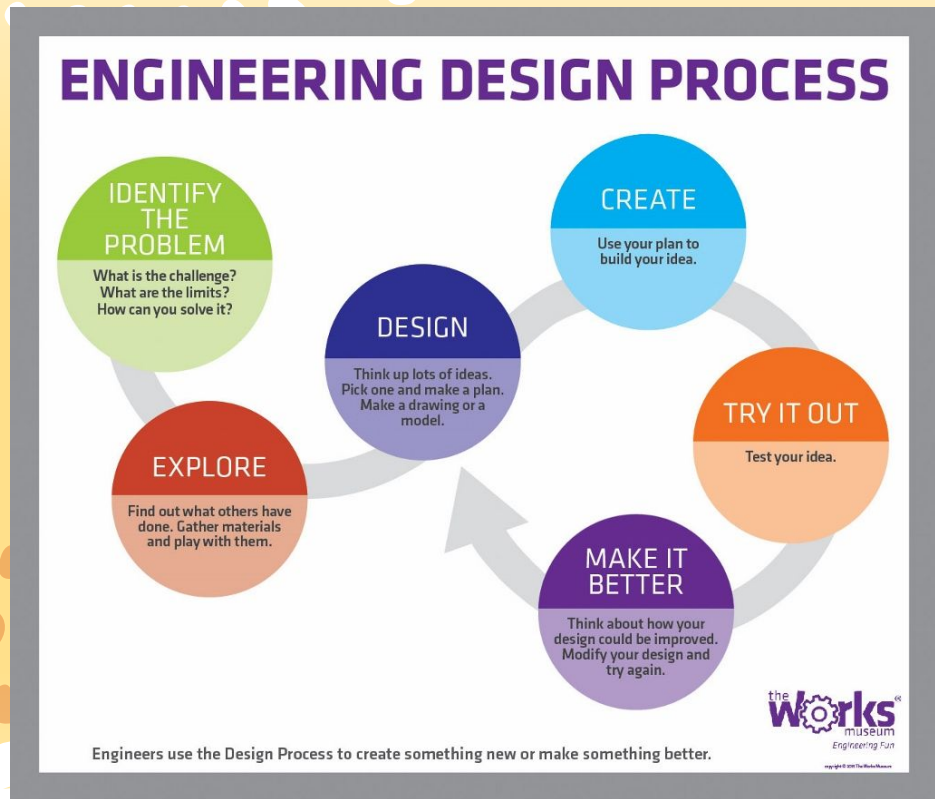


Bad/Flawed Design

Dieter Rams' Principles of Good Design

- ☐ Good Design Is **Innovative**
- ☐ Good Design **Makes a Product Useful**
- ☐ Good Design Is **Aesthetic**
- ☐ Good Design **Makes A Product Understandable**
- ☐ Good Design Is **Unobtrusive**
- ☐ Good Design Is **Honest**
- ☐ Good Design Is **Long-lasting**
- ☐ Good Design Is **Thorough Down to the Last Detail**
- ☐ Good Design Is **Environmentally Friendly**
- ☐ Good Design Is as **Little Design as Possible**

Engineering Design Process



Problem Formulation

Recognizing the Need:

- **Identify the Client:**

Do we know who this is

Do they know what their “need” is

- **Gathering the Client Needs:**

Interviews

Questionnaires

Focus Group

Be the customer

- **Recognizing the Need:**

Rules for interacting with the client:

1. Keep a broad perspective

2. The client may pose a Potential

Solution and not their original

may lose scope of that need and

direction from

need. Hence, as a designer, you

head in the wrong

the start.

3. Always ask why when someone

says “I need”

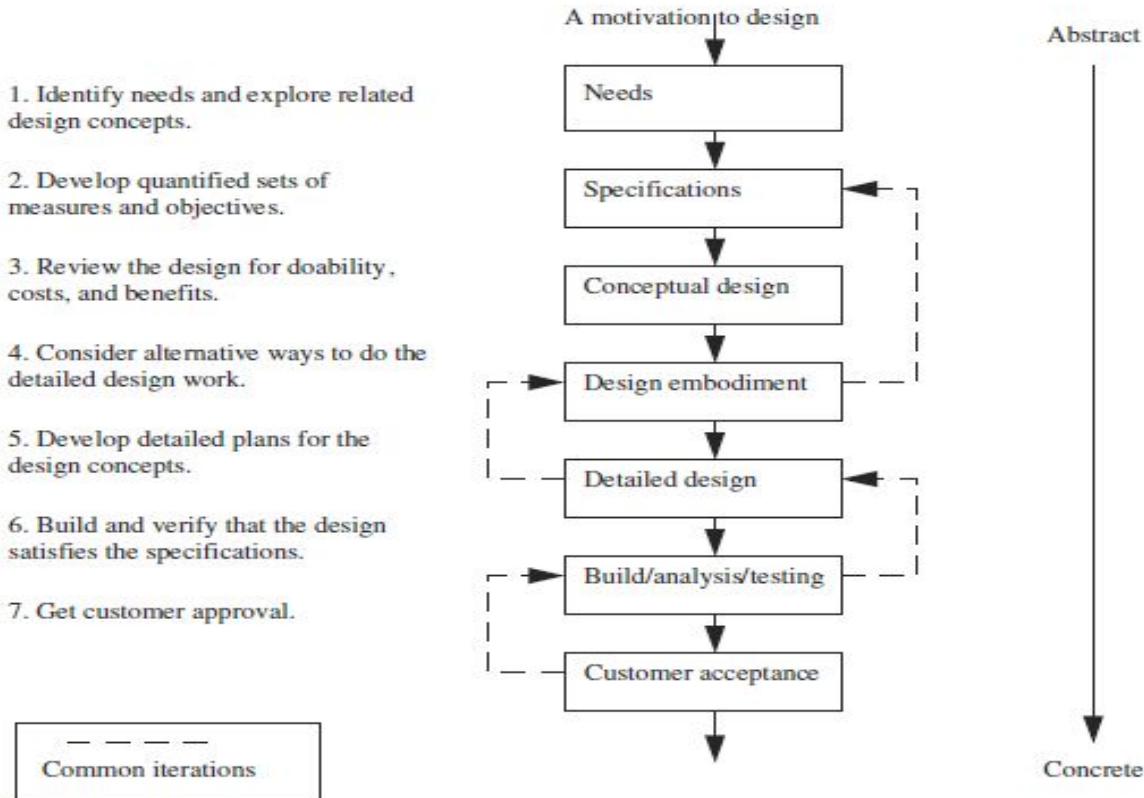
Problem Formulation

Recognizing the Need:

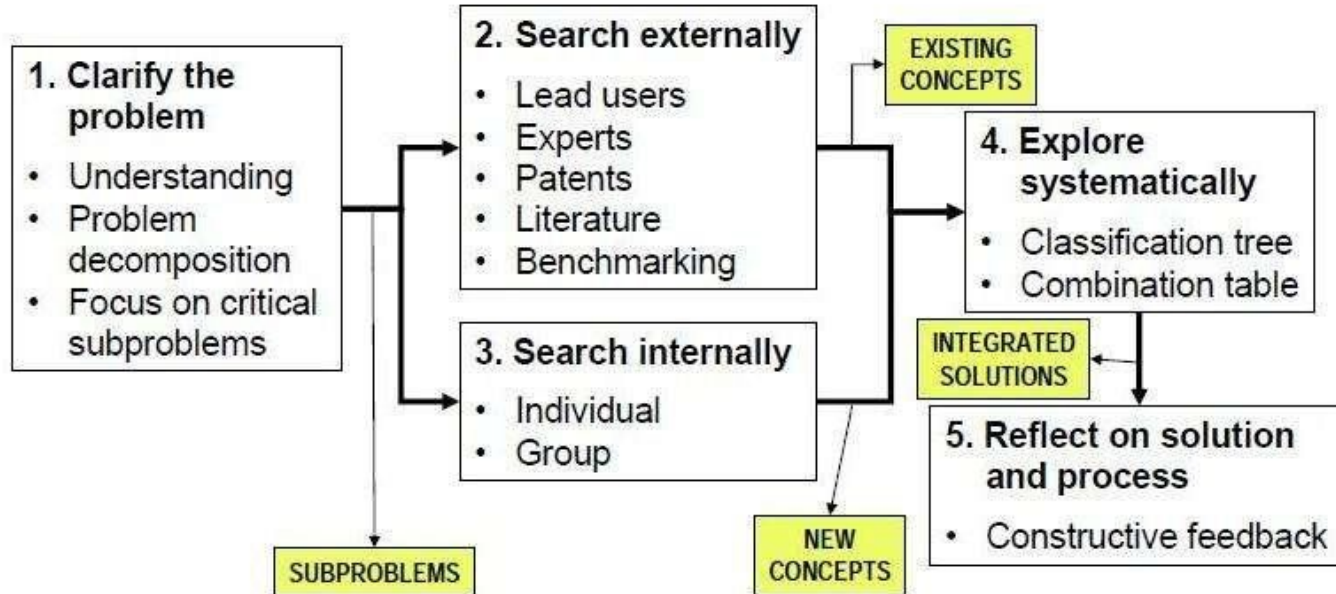
- **Defining the goal:**
Goal Statement
Carefully consider the scope/boundaries of the Goal Statement



A seven-step model of project work.



Concept Generation Process



Project Planning & Management

The project manager is responsible for creating, tracking, and revising the plan. However, the process of planning should include multiple groups and people, typically called stakeholders, or sometimes the audience.

The planning process has multiple steps including

- (1) Breaking the project into a set of tasks,
- (2) Developing a list of approvals and milestones,
- (3) Itemizing required resources,
- (4) Checking the plan for consistency.

The plan often includes a schedule for tasks and time, a budget for money, and a communication list.

Design Categories

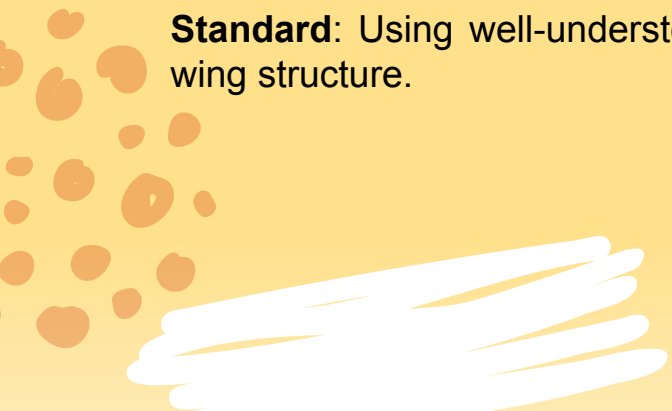


Revolutionary: The design is completely open ended. Concepts are system wide; e.g., number of wings.



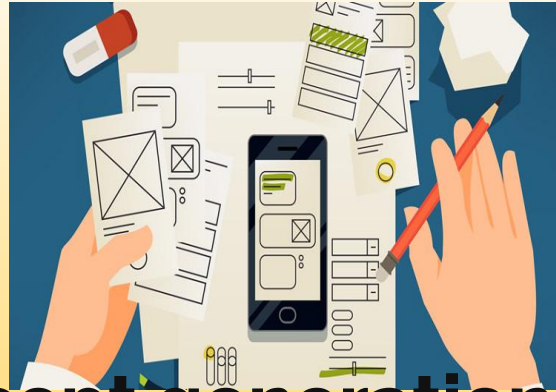
Evolutionary: An existing design is to be refined. Some new concepts are required; e.g., wing geometry.

Standard: Using well-understood approaches to design. The focus is on the details; e.g., wing structure.





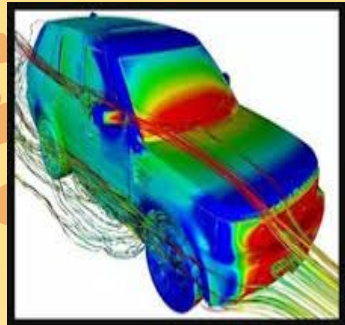
Brainstorming



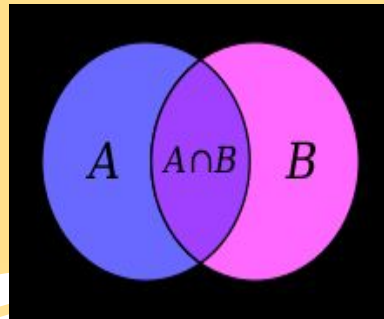
Prototyping



Pilot testing



Simulation



Axiomatic



Ideation

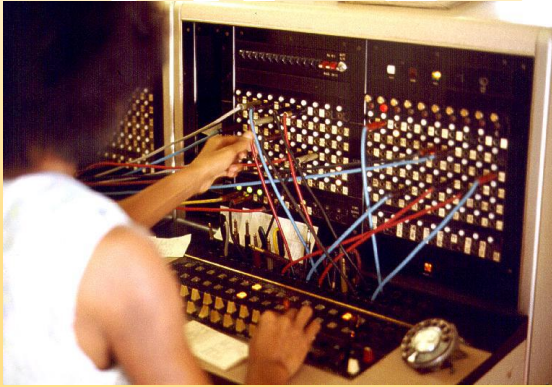


Survey

Major concept generation techniques

The background is a solid light orange color. It features several abstract patterns: a cluster of small orange dashes in the top left, a white scribbled circle in the middle left, a black scribbled circle in the top right, a cluster of black dots in the bottom right, and some orange brushstrokes in the bottom right corner.

Product Design



Technology
Centred Design

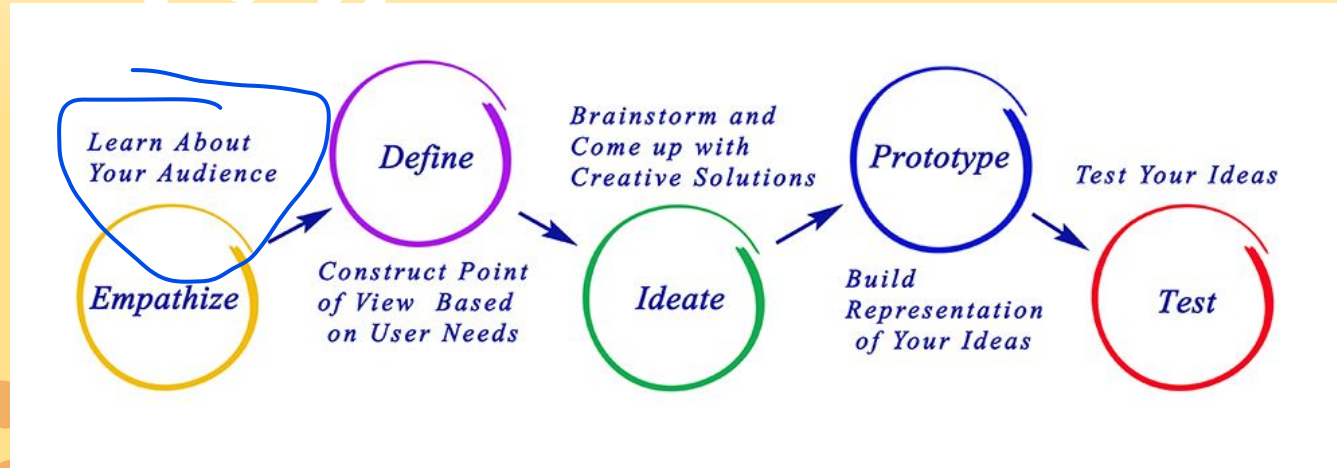


Business
Centred Design

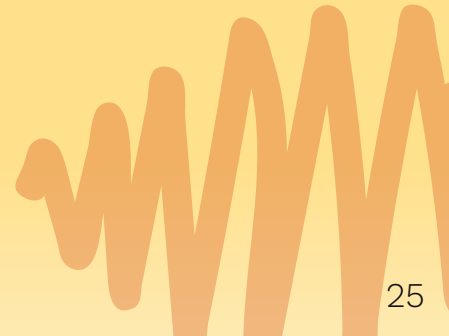
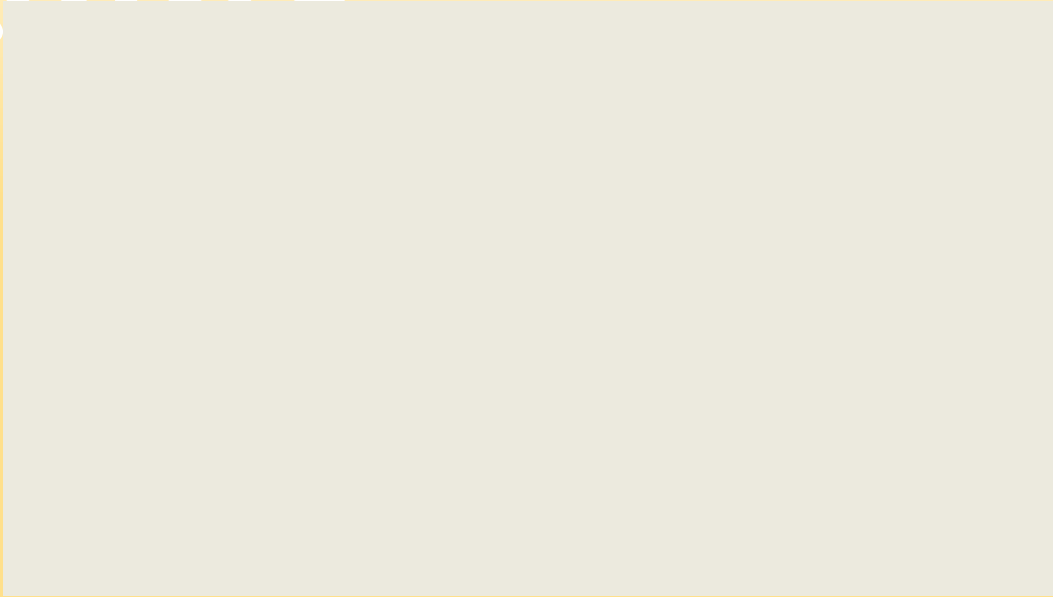


Designer
Centred Design

Design Thinking Process

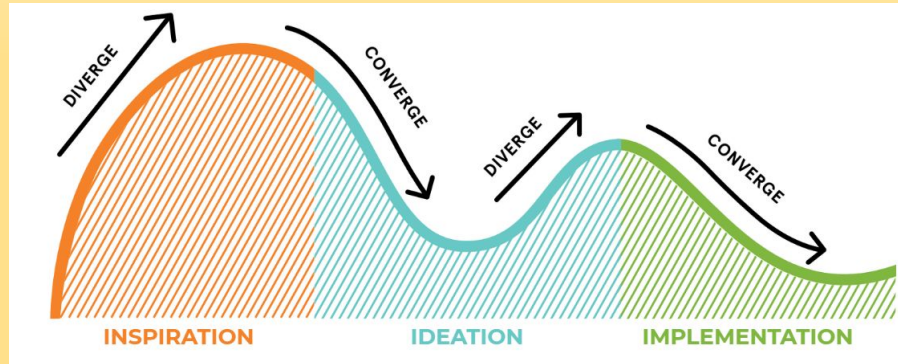


Human Centred Design



“Design Thinking” and “Human-Centred Design”

Design Thinking, popularized by Stanford's d.school, is a process that you go through to create solutions that will actually be adopted by people



Human-Centered Design, popularized by IDEO.org, is a mindset that overlays design thinking to ensure that the products are actually relevant and beneficial— in the long run — for the people they are intended to serve.

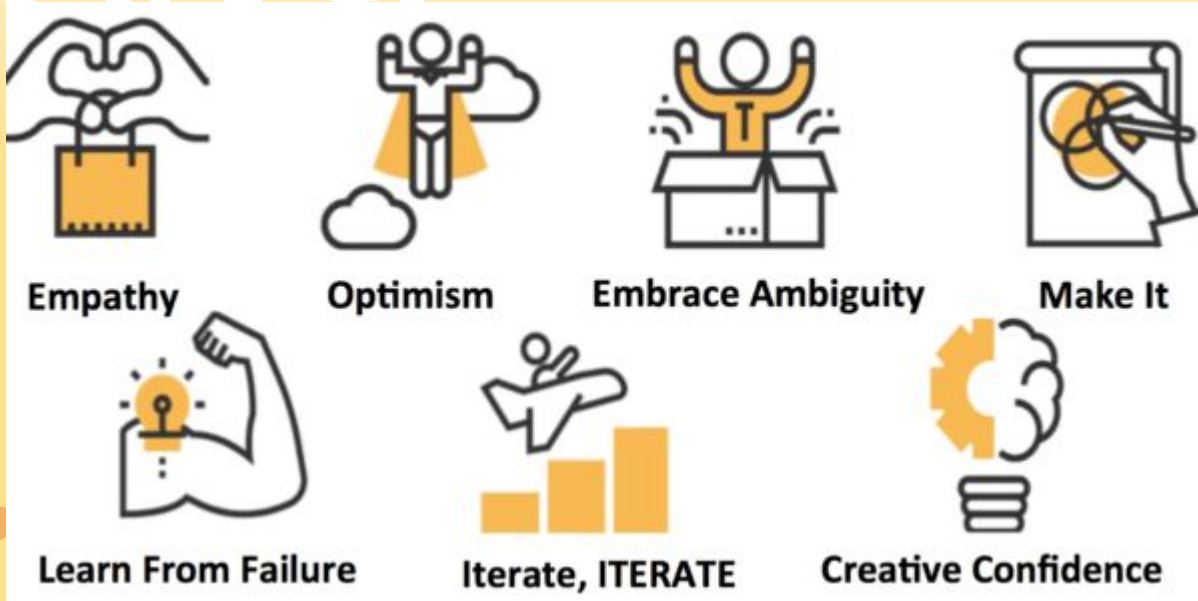
“Design Thinking” and “Human-Centred Design”

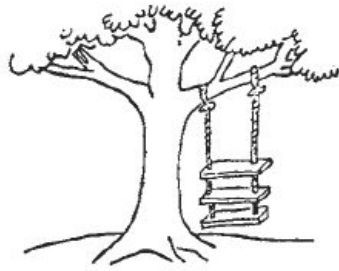


To make this more clear, any business can use *Design Thinking* to build a solution that is capable of making money. For example, a company may use Design Thinking to create a video game or TV show for kids. Applying *Human-Centered Design* on top of this will ensure that the show actually serves the needs of the people watching it (for example meeting the learning objectives of the children watching the show or playing the game)

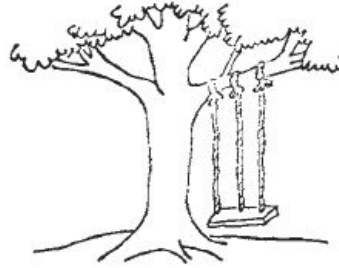


Design Thinking Mindset

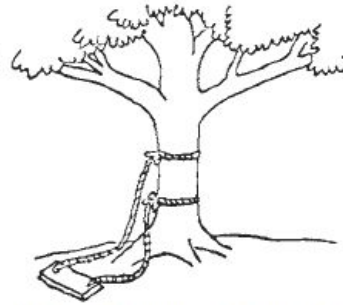




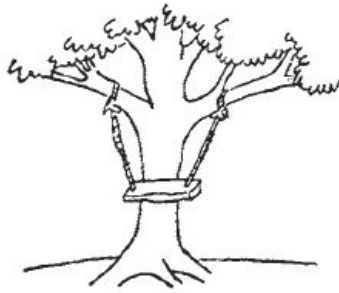
As proposed by the project sponsor



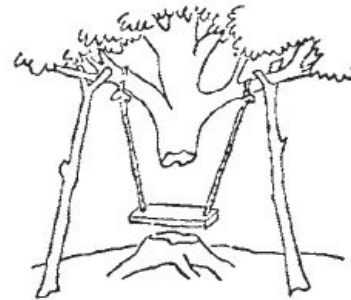
As specified in the project request



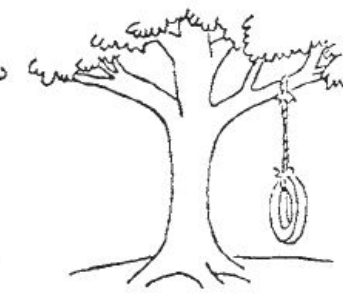
As designed by the senior designer



As produced by manufacturing



As installed at the user's site

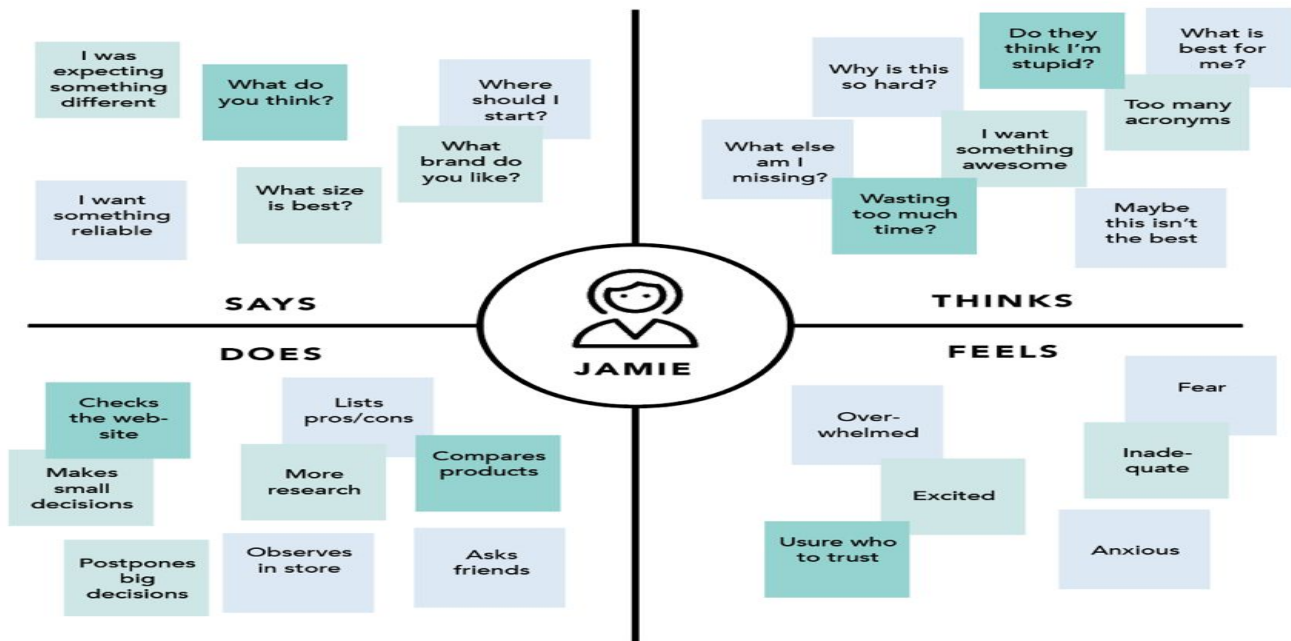


What the user wanted

Design depends on the viewpoint of the individual who defines the problem.

Empathy Map

EMPATHY MAP *Example (Buying a TV)*





Tools for Design Thinking

SCAMPER

S

SUBSTITUTE:

Replace a thing, or concept with something else.



C

COMBINE:

Unite! What? Who? Ideas? Materials?



A

ADAPT:

Adjust to a new purpose. Re-shape? Tune-up?



M

MODIFY, MAGNIFY, MINIFY

Change the colour, sound, motion form, size.

Make it larger, stronger, thicker, higher, longer.

Make it smaller, lighter, slower, less frequent, reduce.



P

PUT TO ANOTHER USE:

Change when, where, location, time, or how to use it.



E

ELIMINATE:

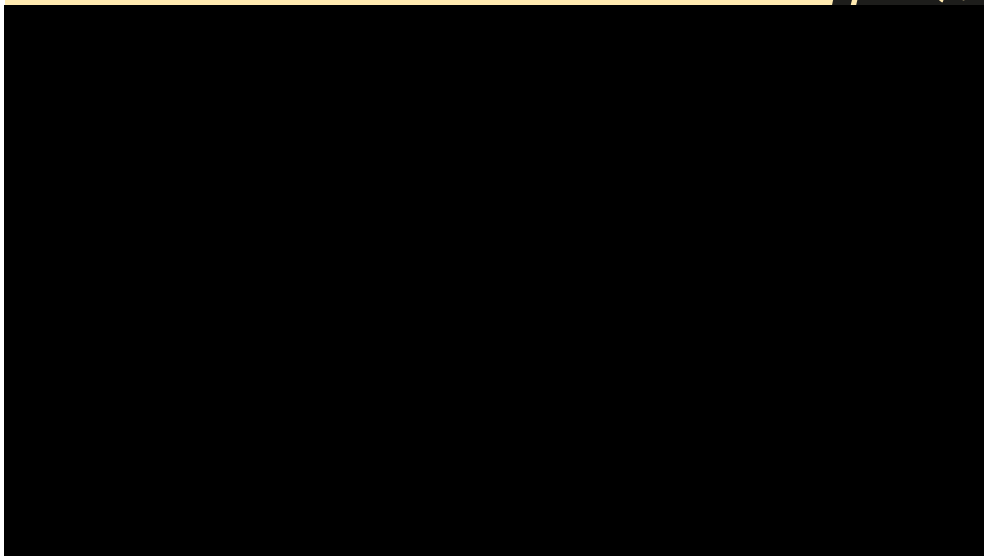
Omit, get rid of, cut out, simplify, weed out...



R

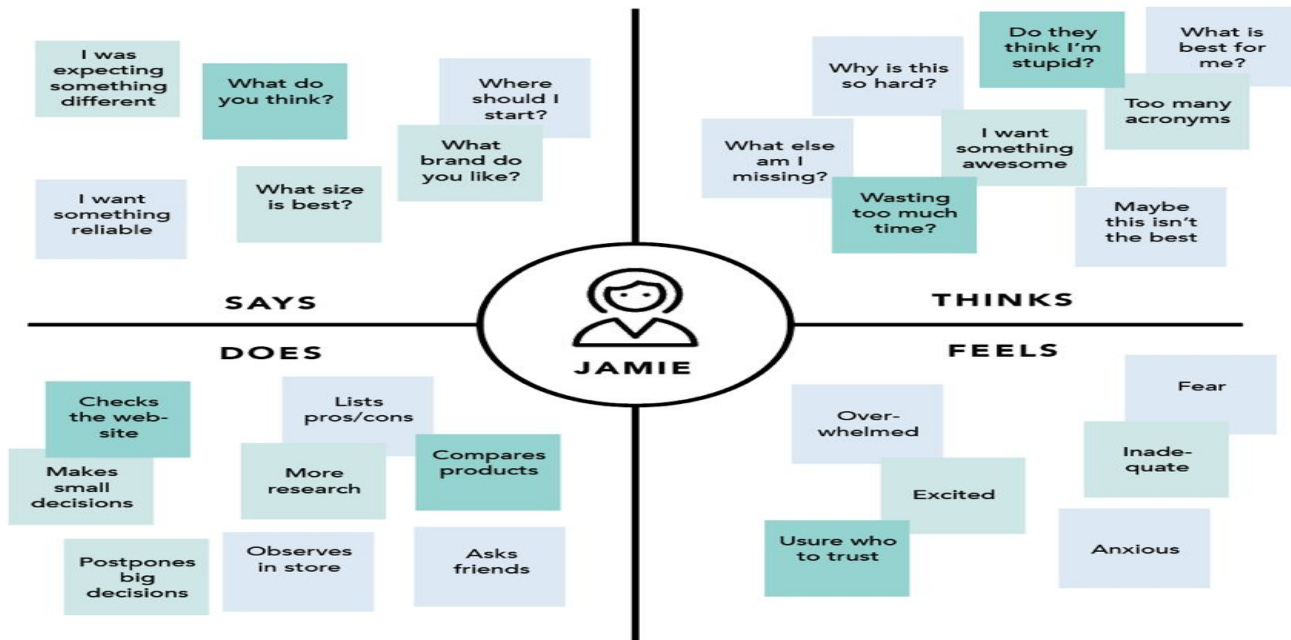
REARRANGE, REVERSE

Change the order, sequence, pattern, layout, plan, scheme, regroup, redistribute...



Empathy Map

EMPATHY MAP *Example (Buying a TV)*



“Mind mapping makes visual thinking come alive



Mind
mapping

Innovation Flowchart

STAGE	SPECIALIST SKILLS REQUIRED	EXAMPLE ACTIVITIES & TOOLS	RISK LEVEL AND HANDLING	FINANCE REQUIRED	KINDS OF EVIDENCE GENERATED	GOAL
1 Exploring opportunities & challenges	→	→	→	→	→	→
2 Generating ideas	→	→	→	→	→	→
3 Developing & testing	→	→	→	→	→	→
4 Making the case	→	→	→	→	→	→
5 Delivering and implementing	→	→	→	→	→	→
6 Growing, scaling and spreading	→	→	→	→	→	→
7 Changing systems	→	→	→	→	→	→

Open this link for more information: <https://youtu.be/572wbk1rUAA>



Thanks!